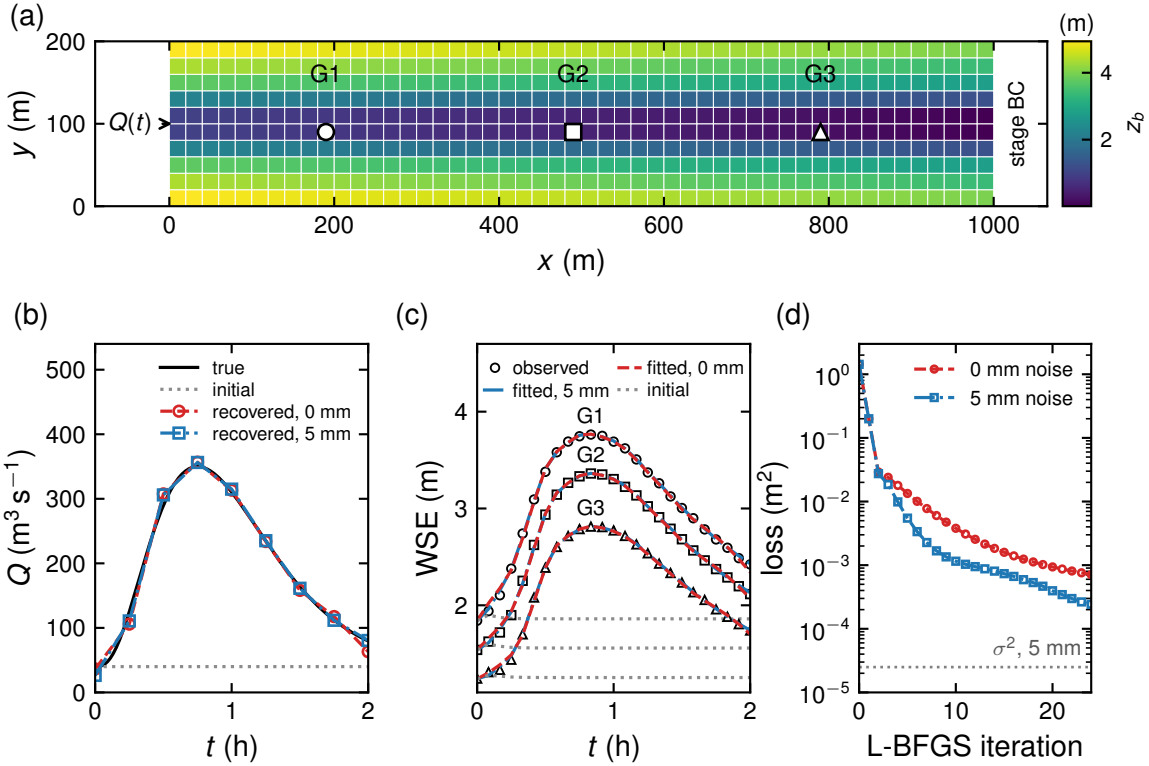

(a)
(m)
200
100
0
y (m)
Q(t)
G1
G2
G3
stage BC
4
2
z_b
0
200
400
600
800
1000
x (m)
(b)
true
initial
recovered, 0 mm
recovered, 5 mm
500
400
300
200
100
0
Q ($m^3 s^{-1}$)
0
1
2
t (h)
(c)
observed
fitted, 0 mm
fitted, 5 mm
initial
G1
G2
G3
4
3
2
WSE (m)
0
1
2
t (h)
(d)
0 mm noise
5 mm noise
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
loss (m^2)
σ^2, 5 mm
0
10
20
L-BFGS iteration